

0.1 Section 2.2 - Set Operations

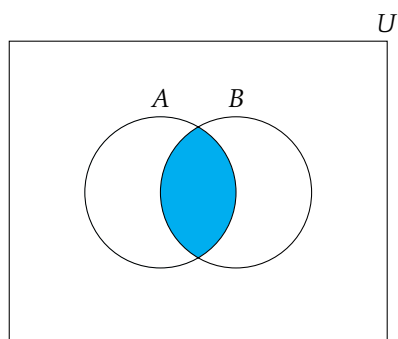
Let's consider a venn diagram which is a way of visualizing sets

Let U represent the universal set

Definition 1 (Intersection)

Elements that are in **both** A and B is called the intersection of A and B

$$A \cap B$$



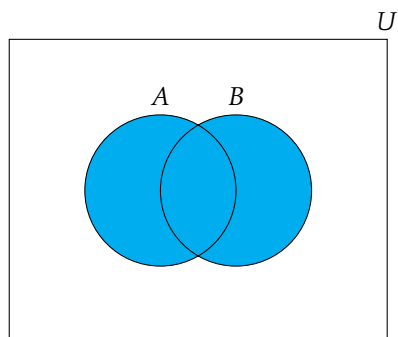
Definition 2 (Union)

Elements that are in **either** A or B is called the union of A and B

$$A \cup B$$

Also

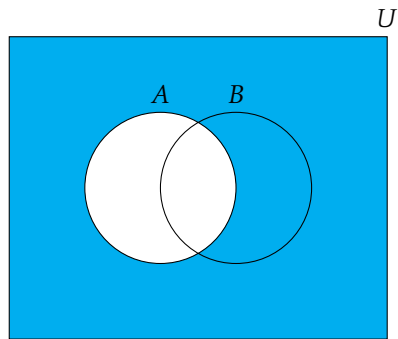
$$|A \cup B| = |A| + |B| - |A \cap B|$$



Definition 3 (Compliment)

Elements that just aren't in A is called the **compliment** of A

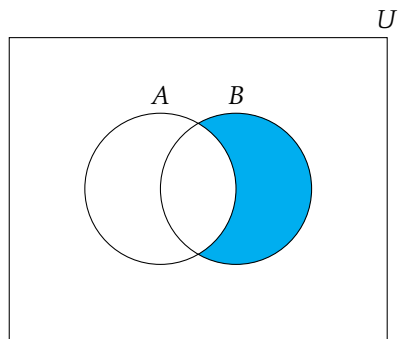
$$\bar{A}$$



Definition 4 (Difference)

Elements in B but not in A is called the **difference**

$$B - A$$



With sets, we have 3 operations:

1. Compliment $\bar{A}$ (analogous to $\neg$)
2. Union $A \cup B$ (analogous to $\vee$)
3. Intersection $A \cap B$ (analogous to $\wedge$)

Example 1

Let

$$\begin{aligned}
 U &= \{x \in \mathbb{Z} \mid 0 \leq x \leq 20\} \\
 A &= \{x \in U \mid x \text{ is even}\} \\
 B &= \{x \in U \mid x \text{ is divisible by } 3\}
 \end{aligned}$$

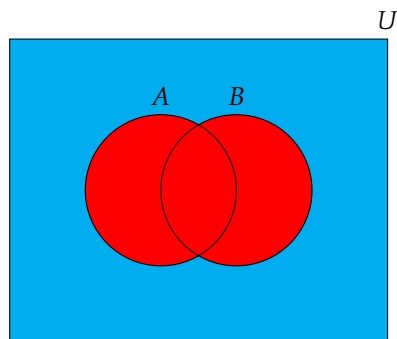
Determine the following

1. $A = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$
2. $B = \{0, 3, 6, 9, 12, 15, 18\}$
3. $A \cap B = \{0, 6, 12, 18\}$ (what they have in common)
4. $A \cup B = \{0, 2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$ (whats in either one)
5. $\overline{A} = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$ (whats not in A)
6. $B - A = \{3, 9, 15\}$ (what's in B thats not in A)
7. $|A| = 11$ (number of elements in A)
8. $|B| = 7$ (number of elements in B)
9. $|A \cup B| = 14$
10. $|A \cap B| = 4$ (number of elements that A and B have in common)
11. $|A| + |B| - |A \cap B| = 11 + 7 - 4 = 14 = |A \cup B|$

But how do these operations interact together?

Example 2

- $\overline{A \cup B}$
- $\text{Red} = A \cup B$
 $\text{Blue} = \overline{A \cup B}$

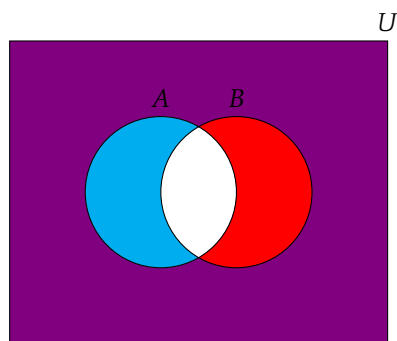


- $\overline{A \cup B}$

Red = $\overline{A}$

Blue = $\overline{B}$

Purple = $\overline{A \cap B}$



$$\therefore \overline{A \cap B} = \overline{A} \cup \overline{B}$$

1. DeMorgan's Laws

- $\overline{A \cup B} = \overline{A} \cap \overline{B}$
- $\overline{A \cap B} = \overline{A} \cup \overline{B}$

2. Distributive Laws

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

3. Complementation Law

- $\overline{\overline{A}} = A$

4. Commutative Law

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$